# DSCI 445 Final Project
# Group 3

# Chosen Problem

- “Store Sales - Time Series Forecasting” From Kaggle
- Goal: Test the capability of different models to accurately forecast store sales based on past yearly data.
- Evaluation Metric for Competition: Root Mean Squared Logarithmic Error

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\log(1+\hat{y_i})-\log(1+y_i)\right)^2}$$

- $n$ is the total number of instances,
- $\hat{y_i}$ is the predicted value of the target for instance (i),
- $y_i$ is the actual value of the target for instance (i), and,
- $\log$ is the natural logarithm.

- Competition Link: Store Sales Time Series Forecasting

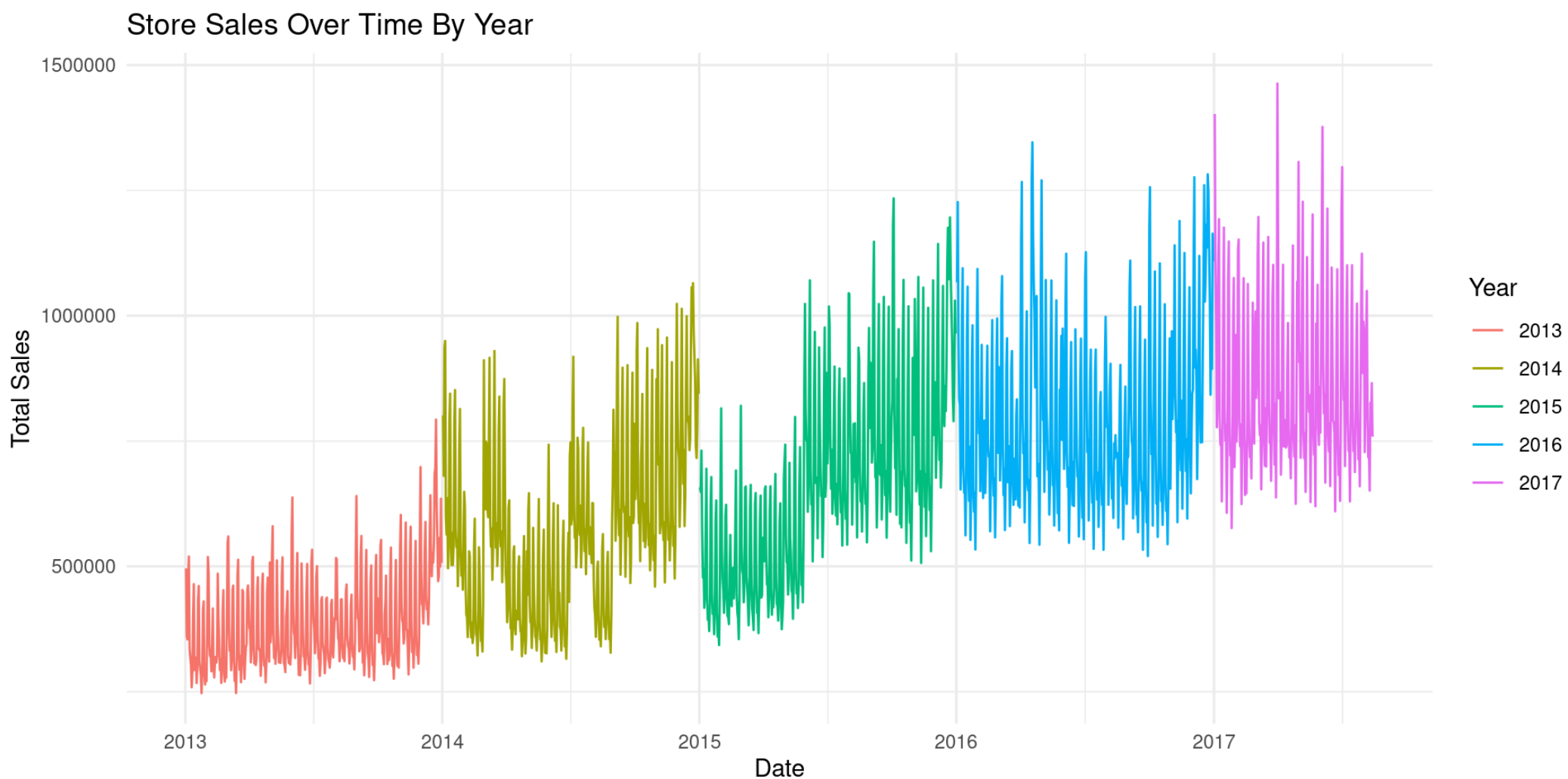
# Description of Data

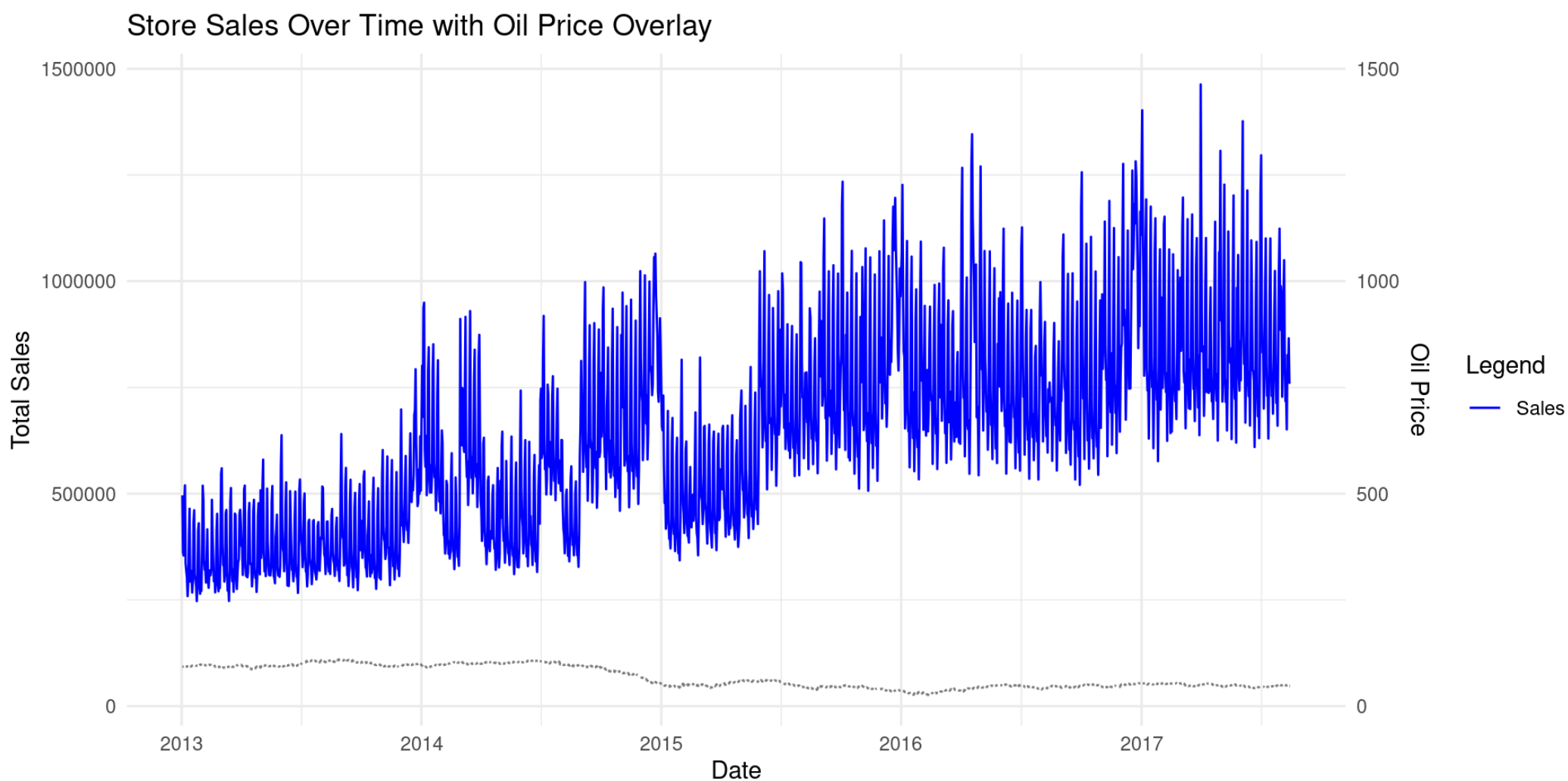
- Given Files: 1 train, 1 test, 4 additional datasets
  - `train.csv`, `test.csv`
  - `stores.csv`
  - `oil.csv`
  - `transactions.csv`
  - `holidays_events.csv`
- Training Data: 3,000,888 rows, 16 columns (prior to cleaning & transformations)
- **Target Feature**: `sales`

# Time Series Considerations

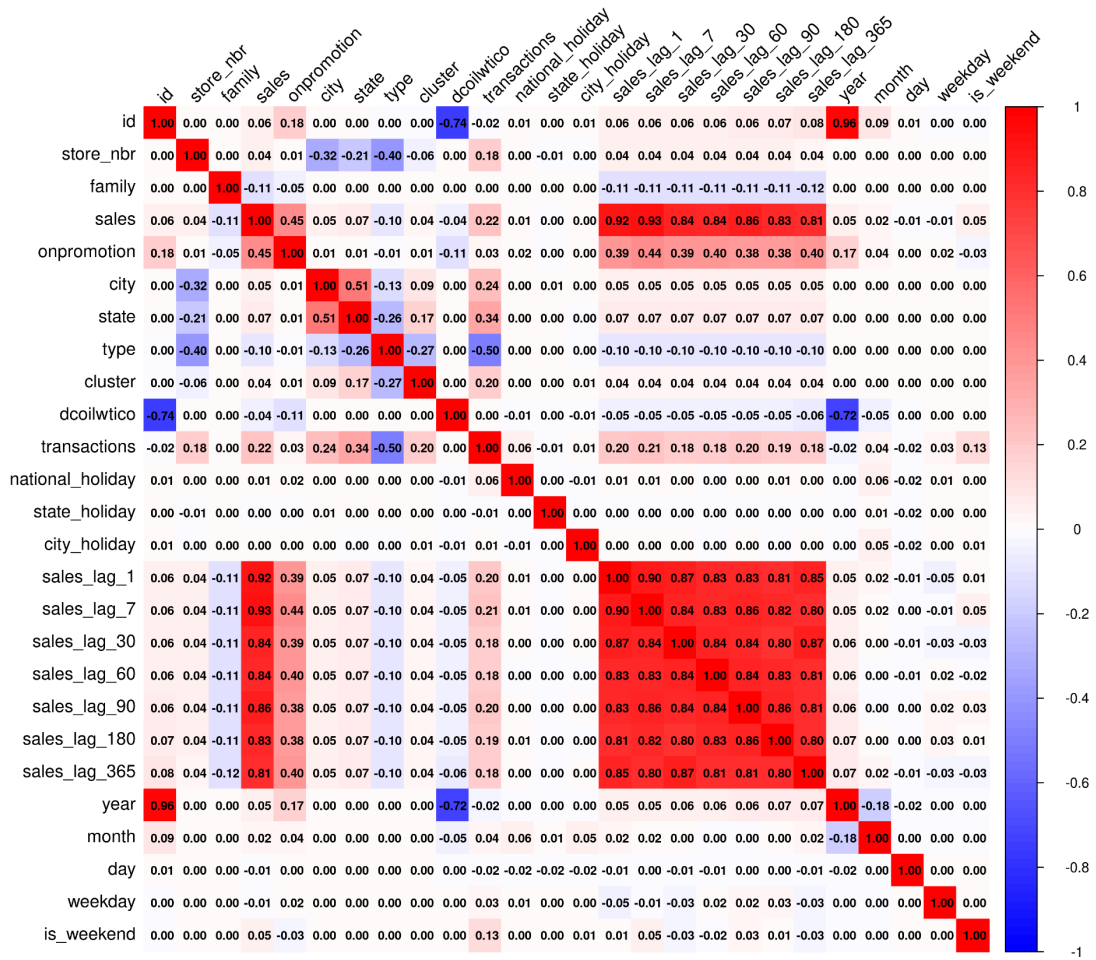
- **Lag Variable**: Represents the value from a previous point in time
  - Captures time dependencies and relationships to past observations
- Engineered 7 lag features for `sales`: 1, 7, 30, 60, 90, 180, 365 (units in days)

# Exploratory Data Analysis




Store Sales Over Time with Oil Price Overlay
1500000
1000000
500000
0
Total Sales
1500
1000
500
0
Oil Price
2013
2014
2015
2016
2017
Date
Legend
Sales

# Correlation Heatmap



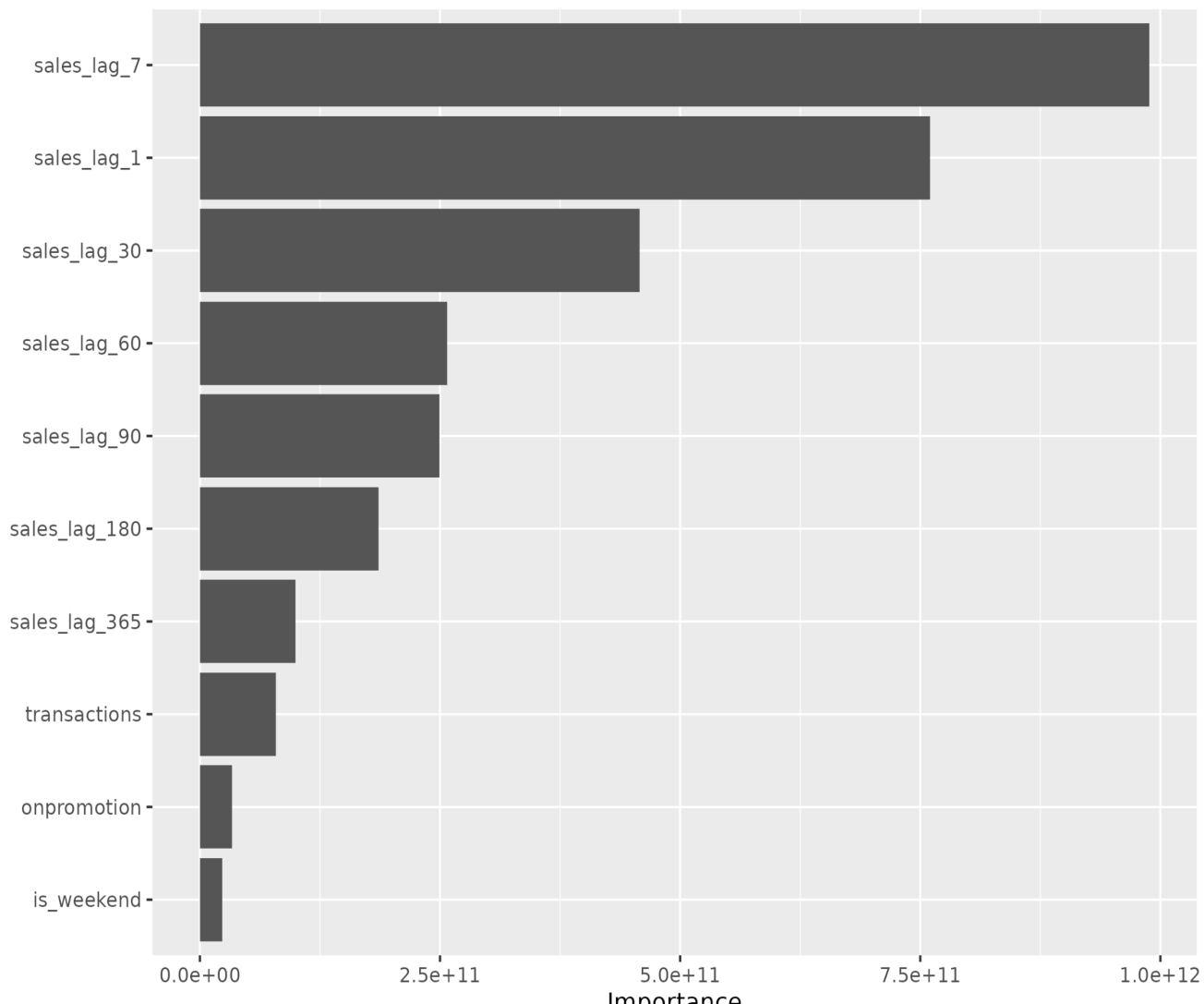
# Lasso Model

- **Regularization**: Shrinks coefficient estimates to 0
  - Lasso's way of performing variable selection
  - Penalty/Lambda: Degree of regularization
    - **Optimal Lambda (CV)**: 1e-04
- **Variable Selection**: 153 predictors -> 85 predictors
  - Large predictor space due to as.factor()
  - Predictor space reduced by almost half
- **RMSLE**: 1.778, **615/742**
- Purpose of lasso is to compare against other models

# Random Forest

- **Accuracy**:
  - Large number of trees average to reduce risk of overfitting
  - Good fit without tuning hyperparameters
- **Feature Selection**:
  - Selects most important predictors for each split
  - Measures predictor importance based on reduction of overall model error
- **Model**: Number of trees: **500**, mtry = **7**, min.node.size = **5**
- $R^2$: **96.03%**
- **OOB RMSE: 236.04**
- **TEST RMSLE: 0.69823, 428/747**

# Feature Selection: Importance Plot



# Reasoning for XGBoost

- XGBoost builds models iteratively, focusing on improving the performance of the previous model by learning from its errors.
- Exceptional ability to handle large datasets.
- Effectively manages complicated nonlinear relationships.

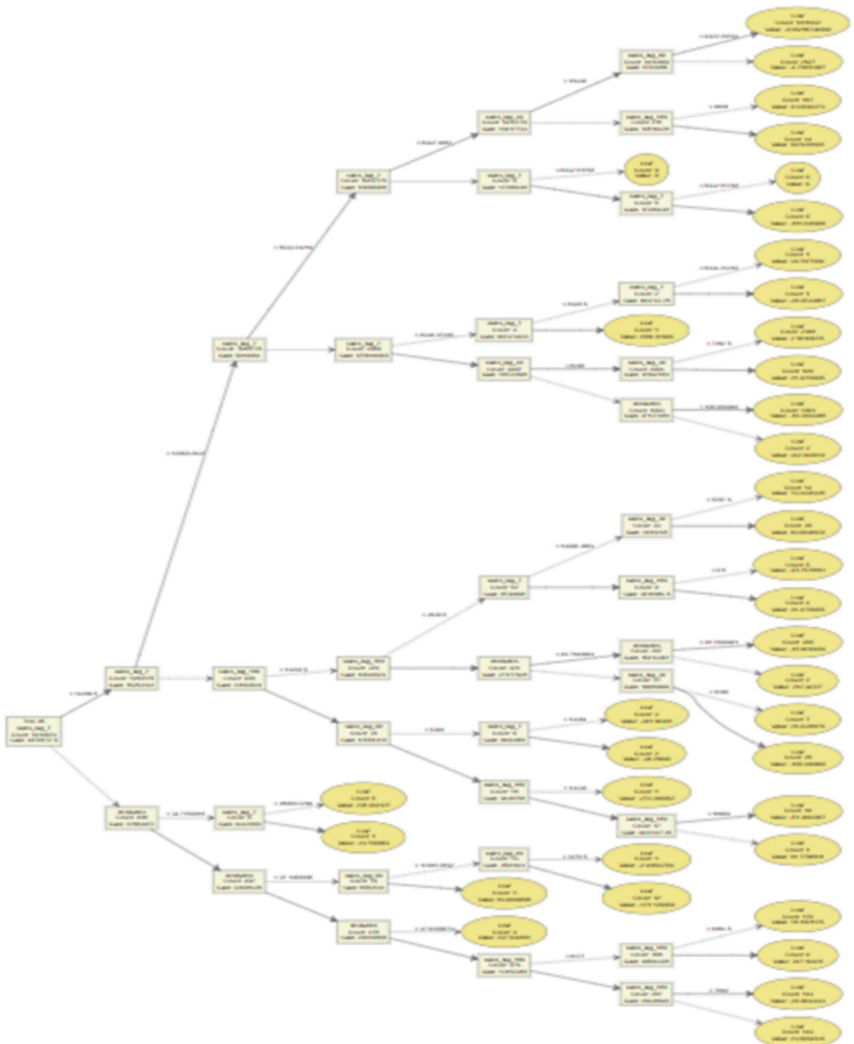
1. **XGBoost's Native Handling**:
    - Automatically recognizes and handles missing values during training.
    - Utilizes a sparsity-aware algorithm to navigate missing data effectively.
2. **Rolling Average Imputation**:
    - Smoothens short-term fluctuations in lagged features.
    - Calculates the mean over a specific window (e.g., 7 days) for imputation.
    - Maintains temporal patterns in the data.

# Training Native Handling Model

```{r}
params <- list(
  objective = "reg:squarederror",
  eval_metric = c("rmse", "mae"),
  max_depth = 6,
  eta = 0.1,
  subsample = 0.8,
  colsample_bytree = 0.8
)

cat("Training with Native Missing Value Handling...\n")
watchlist <- list(train = dtrain, valid = dvalid)
model_native <- xgb.train(
  params = params,
  data = dtrain,
  nrounds = 500,
  watchlist = watchlist,
  early_stopping_rounds = 10,
  verbose = 1
)
```

# XGBoost



# Autoregressive Integrated Moving Average (ARIMA) Model

- Reasoning:
  - Model is specifically used for time series data
  - Could potentially identify more granular patterns within specific time periods
- Able to use lagged values of variables to better quantify autocorrelation within the data set
- Autocorrelation: “The degree of correlation of the same variables between two successive time intervals”

# ARIMA Base and Seasonal Parameters

- Base Function Parameters (5,1,1):
  - 5 - Number of past observations that are used to predict the current value
  - 1 - Number of times the current value is subtracted from previous value
    - Ensures Stationary Data
  - 1 - Number of lagged forecast errors used to predict current value
    - Uses most recent forecast error to adjust prediction
- Seasonal Function Parameters (2,1,2) and Frequency Term (7):
  - 7 - Number of days the seasonal parameters use when predicting
  - 2 - Number of past observations used to predict current value (7, 14 days prior)
  - 1 - Number of times current value is subtracted from value 7 days prior
  - 2 - Number of lagged forecast errors used to predict current values (7, 14 days prior)

# ARIMA Shortcomings

- RMSLE of the ARIMA Model: 3.93254
- Potential reasons for higher evaluation metric:
  - Aggregation of data was required to train model
  - Deaggregation using a standard proportion of total sales was needed for submission
  - This caused potential bias in the amount of sales coming from the “store_nbr” and “family” variables
- Alternative Methods:
  - Implemented Fourier Features to find more granular patterns within the specified frequency
  - Create two separate models that share the entirety of the information in the data set using a Bayesian structure

# Results

Model Comparison

| Model | RMSLE | Ranking |
|---|---|---|
| ARIMA | 3.933 | 716 |
| Lasso | 1.788 | 615 |
| XGBoost | 0.711 | 585 |
| Random Forest | 0.684 | 468 |

# Thank you!

Questions?